BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Autumn 1 AP1 Assessment and Review

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: Answer all four before turning to the answers.

1. What do MAR and MDR each store?

2. State one difference between bubble and insertion sort.

3. What is the purpose of input validation?

4. Calculate raw pixel-data size for a 20 x 10 bitmap at four bits per pixel, then convert to bytes.

Today's objectives

Recall key ideas from Paper 1 and introductory Paper 2. Show careful working in calculations and traces.

Key vocabulary - Read, cover, recall

register
Small, fast CPU storage holding data or addresses.

compression
Reducing bits needed to represent a file.

abstraction
Ignoring irrelevant problem details.

validation
Checking data against specified input rules.

Write one definition in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes. P1 + P2 aims

Learn and practise

New learning - Four key ideas

PAPER 1	Recall CPU, storage, binary, images and sound.
PAPER 2	Recall ADA, flowcharts, searching, sorting and validation.
SHOW WORKING	Write formula, values, bits, then bytes.
REVIEW & IMPROVE	Find one strength and one topic to revisit.

SHOW EACH STEP - IMAGE SIZE CALCULATION

PIXELS	BITS	BYTES
50 x 40 = 2000	2000 x 8 = 16000	16000 / 8 = 2000

For calculations: formula -> working -> unit.

I Do - Worked example

Model a synoptic answer: calculate bitmap storage and explain a suitable device for a portable laptop.

1. 50 x 40 = 2,000 pixels.
2. 2,000 x 8 = 16,000 raw bits.
3. 16,000 / 8 = 2,000 bytes.
4. Choose an SSD for a knocked laptop: no moving heads.
5. Drawback: SSDs often cost more per GB than HDDs.

We Do - Guided practice

Find the raw size of a 5 x 4 bitmap at 8 bits per pixel.

BLACK PEN - complete your explanation:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

Read the OCR Exam Reference Language code before testing it.

```
width = 10
height = 5
depth = 8
print(width * height * depth)
```

1. Predict the raw bitmap size in bits.

2. Change depth to 4 and predict again.

3. Show how to convert the second result to bytes.

You Do - Independent exam questions

Q1. A 100 x 20 image uses 4 bits per pixel. Calculate its raw size in bytes. [3 marks]

Q2. Explain abstraction and give an example of a detail that could be ignored in a storage-capacity calculator. [3 marks]

Q3. A pupil enters a mark of 105 into a program accepting 0-100. Explain one validation rule and one suitable boundary test. [4 marks]

Exit Ticket - True or False?

Answer before checking Page 4.

1. An unsigned 8-bit binary number can represent denary 300. ☐ T ☐ F

2. To turn raw bits into bytes, divide by eight. ☐ T ☐ F

3. An SSD has moving magnetic read/write heads. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes. P1 – P2 remains

Review & improve

GREEN PEN: Mark fairly. Tick correct points and fix any mistakes.

Use the mark allocations; then rewrite one answer in full.

Do Now - Answers

Q1 MAR: address of accessed memory location; MDR: data or instructions being transferred. ☐

Q2 Bubble swaps neighbours; insertion inserts next value into growing sorted section. ☐

Q3 Check data against rules before processing. ☐

Q4 20 x 10 x 4 = 800 bits = 100 bytes. ☐

Code First - Answers

1. 10 x 5 x 8 = 400 bits. ☐

2. With depth 4: 200 bits. ☐

3. 200 / 8 = 25 bytes. ☐

We Do - Check the model

5 x 4 x 8 = 160 bits = 20 bytes.

Exit Ticket - Check

1. FALSE 2. TRUE 3. FALSE

Explain one false statement using the key ideas.

You Do - Self-mark

Q1 100 x 20 = 2000 pixels [1]; x 4 = 8000 bits [1]; / 8 = 1000 bytes [1]. ☐

Q2 Ignore irrelevant detail [1]; retain important detail [1]; device colour need not affect storage [1]. ☐

Q3 Validate mark between 0 and 100 inclusive [2]; test 0 or 100 [1] and accept it [1]. ☐

Try the question again

1. Copy the FULL question you got wrong (or found hardest):

2. In GREEN PEN, write a NEW answer. Include working if needed: